

Total No. of Questions : 4]

SEAT No. :

PE49

[Total No. of Pages : 1

[6579] 351

**T.E. (E & TC) /Electronics(VLSI Design and
Technology)/(Advanced Communication Technology)
MICROCONTROLLERS
(2019 Pattern) (Semester - I) (304184) (Insem)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2 and Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1)** a) Draw and explain internal memory organization of 8051. [5]
b) Explain the port 1 structure of 8051 in detail. [5]
c) State salient features of 8051 microcontroller. [5]

OR

- Q2)** a) Compare Microprocessor and Microcontroller [5]
b) Explain IE register in detail. [5]
c) Enlist various modes of operation of Timer & Explain in details. [5]

- Q3)** a) Draw an interfacing diagram for Stepper motor with 8051. Explain the need of ULN 2003. [7]
b) Describe interfacing of ADC 0808/0809 to 8051 microcontroller with necessary diagram. [8]

OR

- Q4)** a) What are the different types of 7 segment display? Write an embedded C program to display 0 to 9 with a delay of 500ms. [7]
b) Write an embedded C program to blink alternate LEDs connected to PORT 1 of 8051 microcontroller (blink LEDs after 1 sec). [8]

